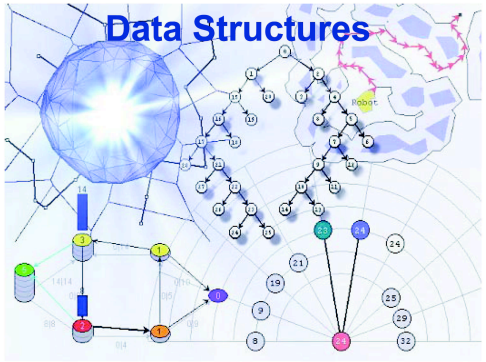
# BBM 201
# DATA STRUCTURES

## Lecture 4:
## Records/Structs and Lists

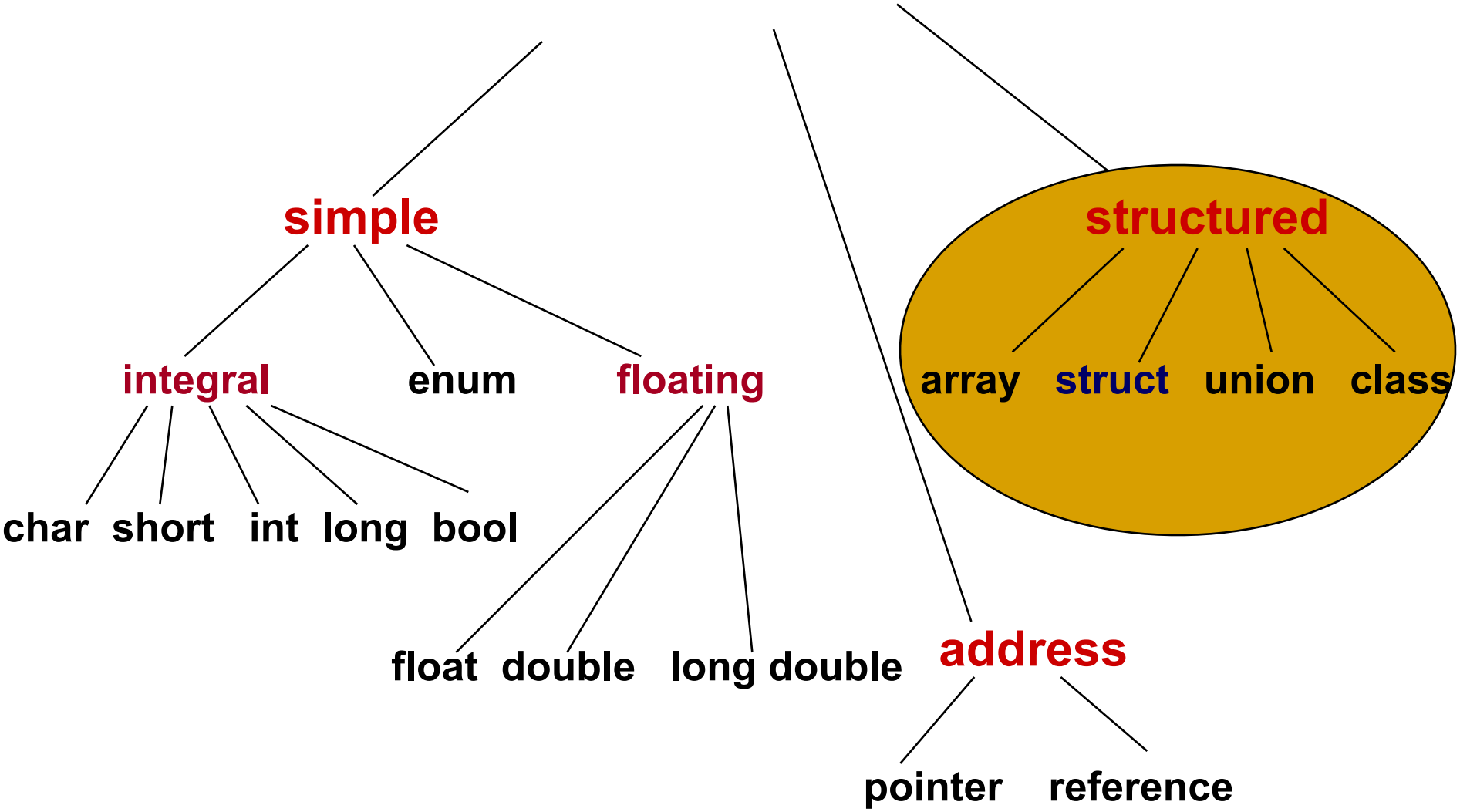


# Objectives

- Learn about records (structs)
- Examine various operations on a struct
- Explore ways to manipulate data using a struct
- Learn about the relationship between a struct and functions
- Discover how arrays are used in a struct
- Learn how to create an array of struct items

- Learn about Lists ADT
- A simple array implementation

# C++ Data Types


simple
integral
enum
floating
char
short
int
long
bool
float
double
long double
address
pointer
reference
structured
array
struct
union
class

# C++ Data Types

- There are simple data types that hold only one value
- There are structured data types that hold multiple values
- The array was the first example of a structured data type that can hold multiple values
- The structure is the second example

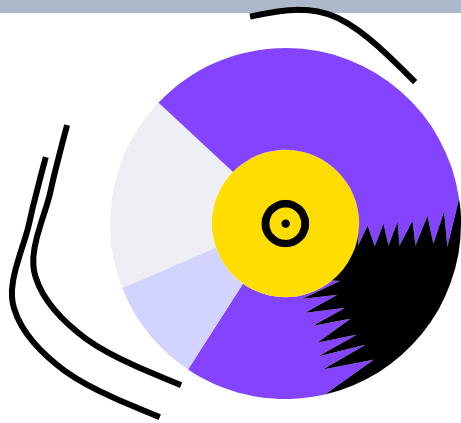
# Structured Data Type

A structured data type is a type in which each value is a collection of component items.

- the entire collection has a single name
- each component can be accessed individually

# Records
# (C++ Structs)

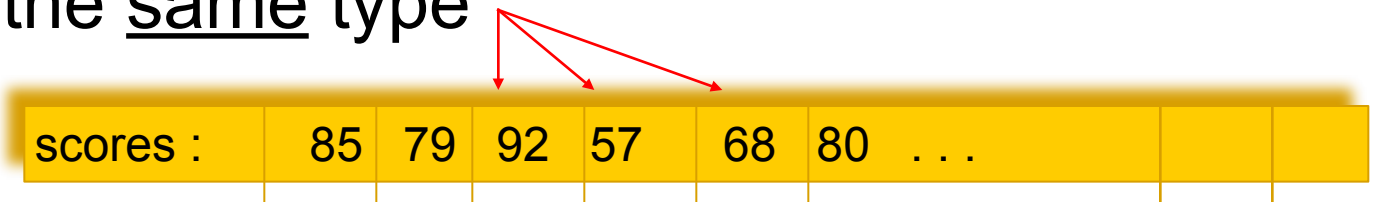
# What to do with records?



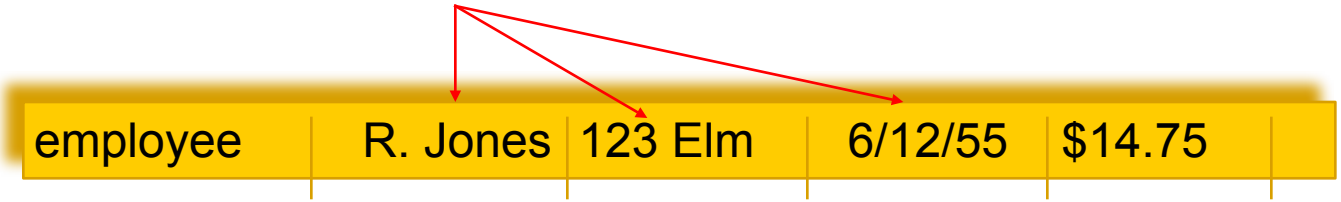
- Declaring records
- Accessing records
- Accessing the field of a record
- Can records be in arrays?

# Records

- Recall that elements of arrays must all be of the <u>same</u> type




- In some situations, we wish to group elements of <u>different</u> types



# Records

- RECORDS are used to group related components of different types
- Components of the record are called <u>fields</u>

| | | | | |
|---|---|---|---|---|
| Employee | R. Jones. | 123 Elm | 6/12/55 | $14.75 |

- In C++
  - record -> struct (structure)
  - fields -> members

# Records

- C++ **struct**
  - structured data type
  - fixed number of components
  - elements accessed by name, not by index
  - components may be of different types

```
struct part_struct {
  char descrip [31], part_num [11];
  float unit price;
  int qty; };
```

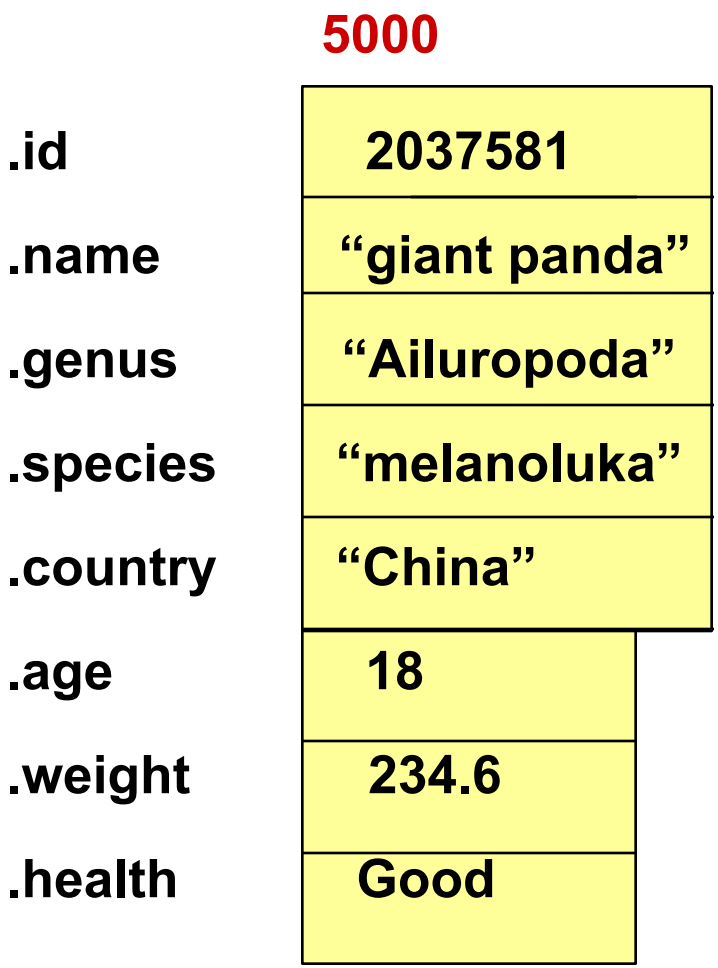
# struct AnimalType

```
struct  AnimalType                    // declares a  struct data type
{                                     //  does not allocate memory
    long        id ;
    string      name ;
    string      genus ;
    string      species ;
    string      country ;
    int         age ;
    float       weight ;
    string      health ;
} ;  // NOTE THE SEMICOLON

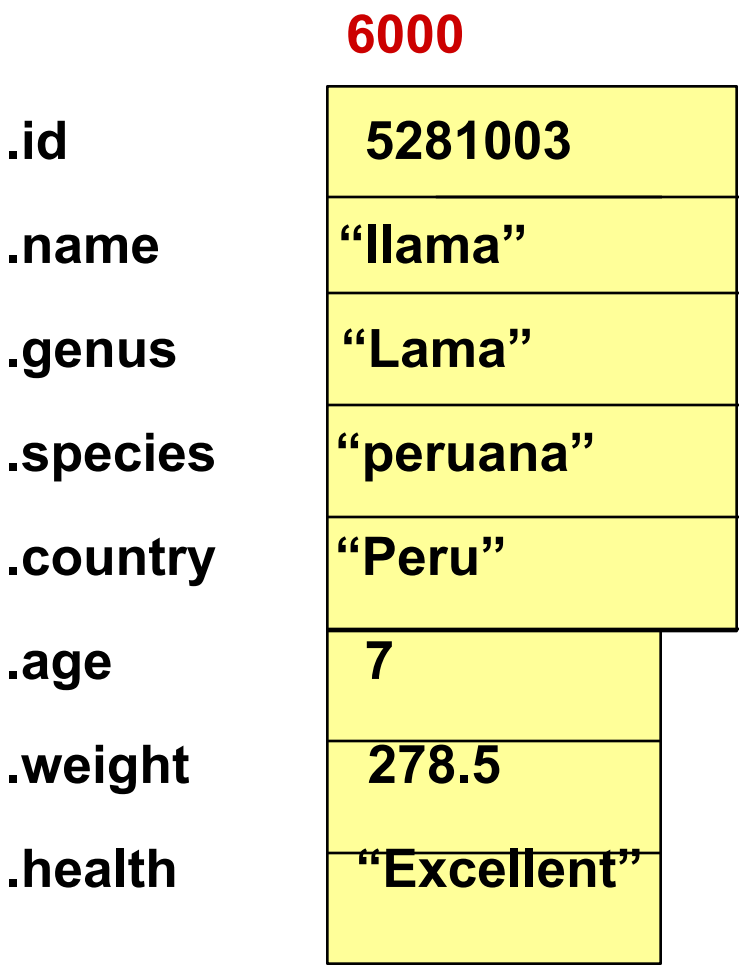
AnimalType    thisAnimal ;            // declare  variables of AnimalType
AnimalType    anotherAnimal ;
```

**struct members**

# thisAnimal


5000
.id
2037581
.name
“giant panda”
.genus
“Ailuropoda”
.species
“melanoluka”
.country
“China”
.age
18
.weight
234.6
.health
Good

# anotherAnimal



# `struct` type Declaration

The struct declaration names a type and names the members of the struct.

It does not allocate memory for any variables of that type!

You still need to declare your struct variables.

# `struct` type declarations

If the struct type declaration precedes all functions it will be visible throughout the rest of the file. If it is placed within a function, only that function can use it.

It is common to place struct type declarations with TypeNames in a (.h) header file and #include that file (more on this later).

It is possible for members of different struct types to have the same identifiers. Also a non-struct variable may have the same identifier as a structure member.

# Accessing `struct` Members

Dot ( period ) is the member selection operator.

After the struct type declaration, the various members can be used in your program only when they are preceded by a struct variable name and a dot.

**EXAMPLES**

**thisAnimal.weight**

**anotherAnimal.country**

# Valid operations on a struct member depend only on its type

```
thisAnimal.age  =  18;

thisAnimal.id    =  2037581;

cin  >>  thisAnimal.weight;

getline ( cin, thisAnimal.species );

thisAnimal.name = “giant panda”;

thisAnimal.genus[ 0 ] = toupper (thisAnimal.genus[ 0 ] ) ;

thisAnimal.age++;
```

# Aggregate Operation

- is an operation on a data structure as a whole, as opposed to an operation on an individual component of the data structure

# Aggregate Operations with Structures

- Limitations on aggregate operations
  - no <u>I/O</u>

```
cout << old_part;
cin >> new_part;
```

  - no <u>arithmetic</u> operations

```
old_part = new_part + old_part;
```

  - no <u>comparisons</u>

```
if (old_part < new_part)
    cout << ..;
```

# Aggregate `struct` Operations

I/O, arithmetic, and comparisons of entire struct variables are NOT ALLOWED!

Operations valid on an entire struct type variable:

- assignment to another struct variable of same type,
- pass to a function as argument (by value or by reference),
- return as value of a function

# Examples of aggregate `struct` operations

```
anotherAnimal   =   thisAnimal ;          // assignment

WriteOut(thisAnimal);                     // value parameter

ChangeWeightAndAge(thisAnimal); // reference parameter

thisAnimal = GetAnimalData( );    // return value of function
```

**NOW WE'LL WRITE FUNCTIONS USED HERE . . .**

```
void  WriteOut(  /*  in  */  AnimalType   thisAnimal)
// Prints out values of all members of thisAnimal
// Precondition:      all members of thisAnimal are assigned
// Postcondition:      all members have been written out
{
    cout   <<  "ID # "  <<  thisAnimal.id  <<  thisAnimal.name  << endl ;

    cout   <<  thisAnimal.genus   <<  thisAnimal.species << endl ;

    cout   <<  thisAnimal.country << endl ;

    cout   <<  thisAnimal.age   << " years "  << endl ;

    cout   <<  thisAnimal.weight << " lbs. "  << endl ;

    cout   << "General health : " ;

    WriteWord ( thisAnimal.health ) ;
}
```

# Passing a `struct` Type by Reference

```
void   ChangeAge ( /* inout */   AnimalType&   thisAnimal )

// Adds 1 to age
// Precondition:      thisAnimal.age is assigned
// Postcondition:     thisAnimal.age == thisAnimal.age@entry + 1

{

    thisAnimal.age++ ;

}
```

```
AnimalType   GetAnimalData ( void )
// Obtains all information about an animal from keyboard
// Postcondition:
//   Function value == AnimalType members entered at  kbd
{
    AnimalType  thisAnimal ;
    char            response ;
    do  {           //  have user enter all members until they are correct
                    .

                    .

                    .
    }  while (response != 'Y' ) ;
    return  thisAnimal ;
}
```

# Hierarchical Structures

The type of a struct member can be another struct type. This is called nested or hierarchical structures.

Hierarchical structures are very useful when there is much detailed information in each record.

**FOR EXAMPLE . . .**

# struct MachineRec

Information about each machine in a shop contains:

an idNumber,

a written description,

the purchase date,

the cost,

and a history (including failure rate, number of days down, and date of last service).

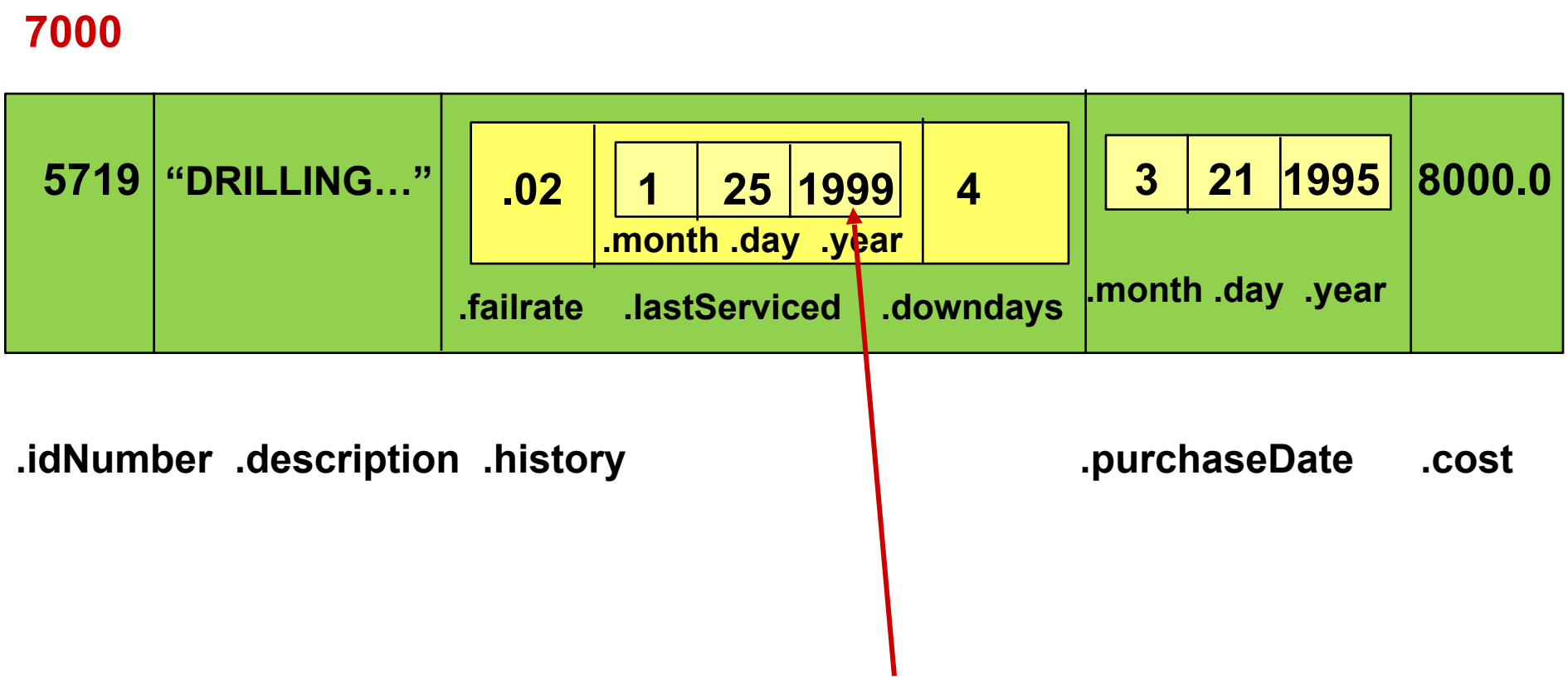
```
struct  DateType
{    int    month ;              //  Assume 1 . . 12
     int    day ;                //  Assume  1 . . 31
     int    year ;               //  Assume 1900 . . 2050
};

struct  StatisticsType
{    float          failRate ;
     DateType       lastServiced ;          //  DateType is a struct type
     int            downDays ;
} ;

struct MachineRec
{    int                idNumber ;
     string    description ;
     StatisticsType   history ;       //  StatisticsType is a struct type
     DateType           purchaseDate ;
     float              cost ;
} ;

MachineRec     machine ;
```

# struct type variable `machine`




**machine.history.lastServiced.year has value 1999**

# Another Struct Example

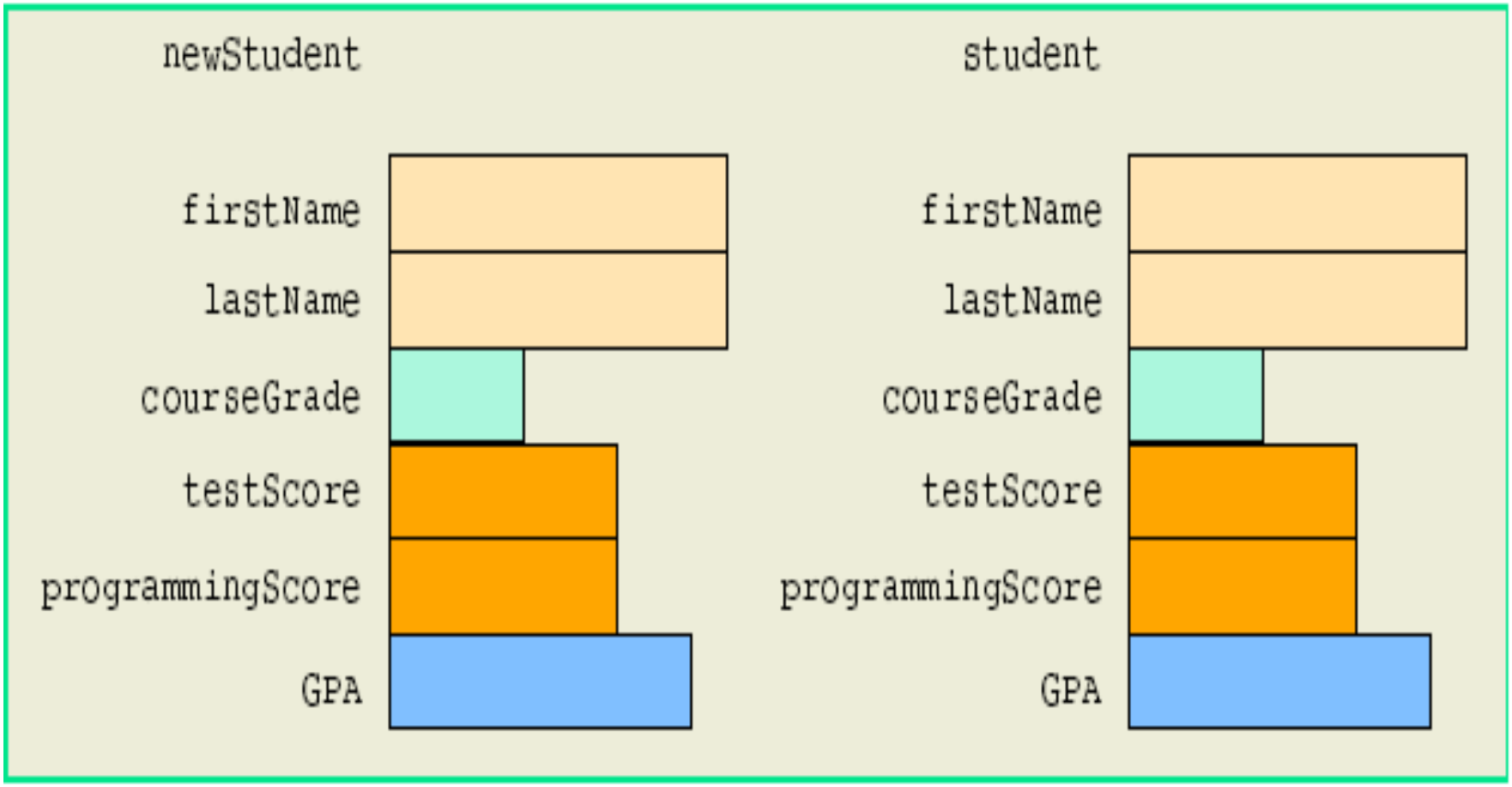
- An example of a studentData struct:

```
struct studentData
{
    string      firstName;
    string      lastName;
    char        courseGrade;
    float       testScore;
    float       programmingScore;
    float       GPA;
};              // NOTE THE SEMICOLON
```

# Declaring a struct

- After you have defined a struct, you can declare variables in your program to be a that struct:

```
studentData  student;
studentData  newStudent;
```




struct newStudent and student

# Assignment

- You can copy one structure to another if they are the same type of

  **student = newStudent;**

- You can copy individual members:

  **newStudent.lastName = student.lastName**

- Or into a variable of the correct type:

  **thisStudentName = student.lastName;**

# Comparison (Relational Operators)

- Compare struct variables member-wise (NOT THE WHOLE STRUCTURE)
- To compare the values of student and newStudent:

```
if(student.firstName == newStudent.firstName &&
  student.lastName == newStudent.lastName)
      .
      .
      .
```

# Input/Output

- No aggregate input/output operations on a struct variable
- Data in a struct variable must be read one member at a time
- The contents of a struct variable must be written one member at a time

# struct Variables and Functions

- A struct variable can be passed as a parameter by value or by reference
- A function can return a value of type struct

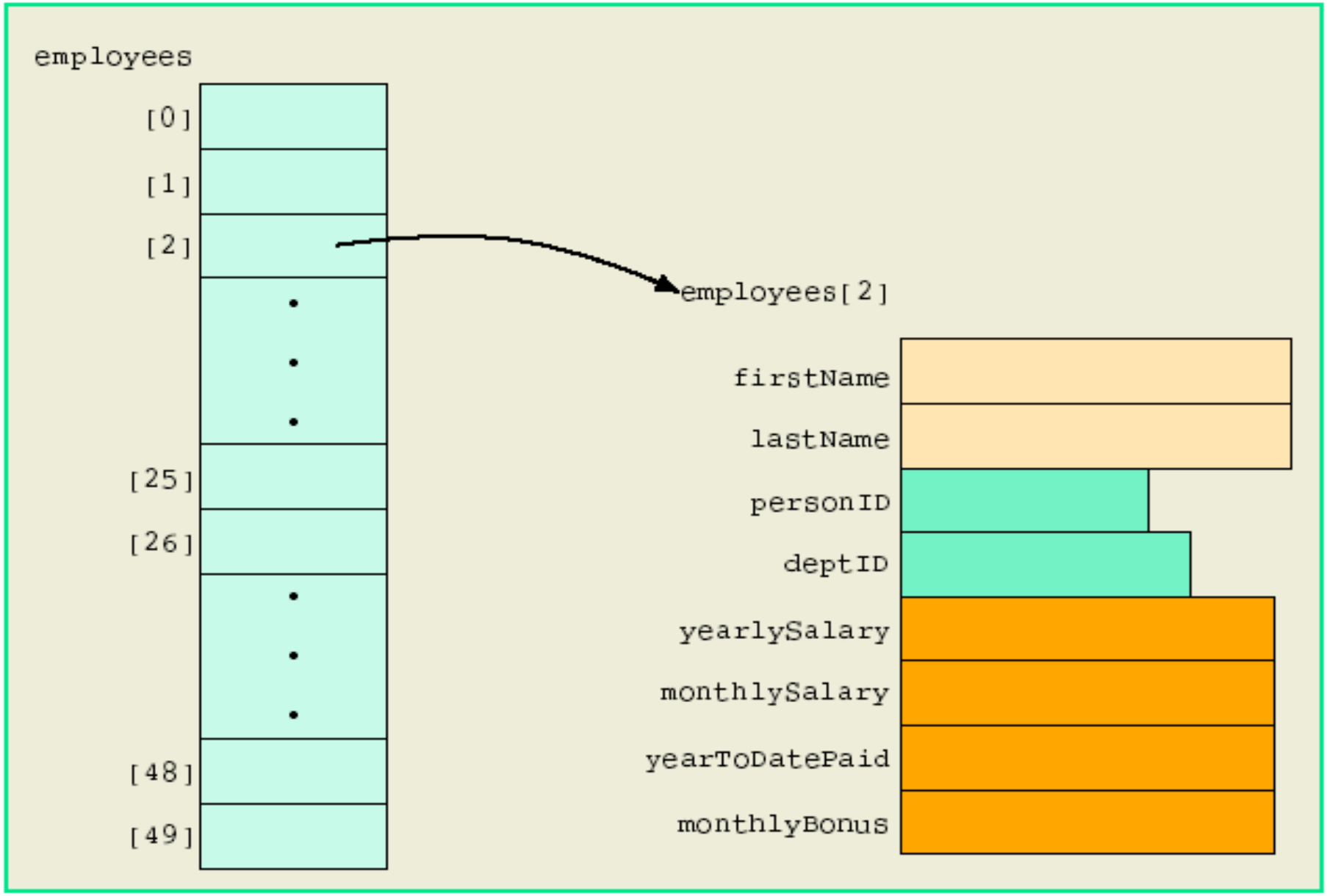
## Arrays vs. Structs

| | Aggregate Operation | Array | `Struct` |
|---|---|---|---|
| 1 | Arithmetic | No | No |
| 2 | Assignment | No | Yes |
| 3 | Input/output | No (except strings) | No |
| 4 | Comparison | No | No |
| 5 | Parameter passing | By reference only | By value or by reference |
| 6 | Function returning a value | No | Yes |

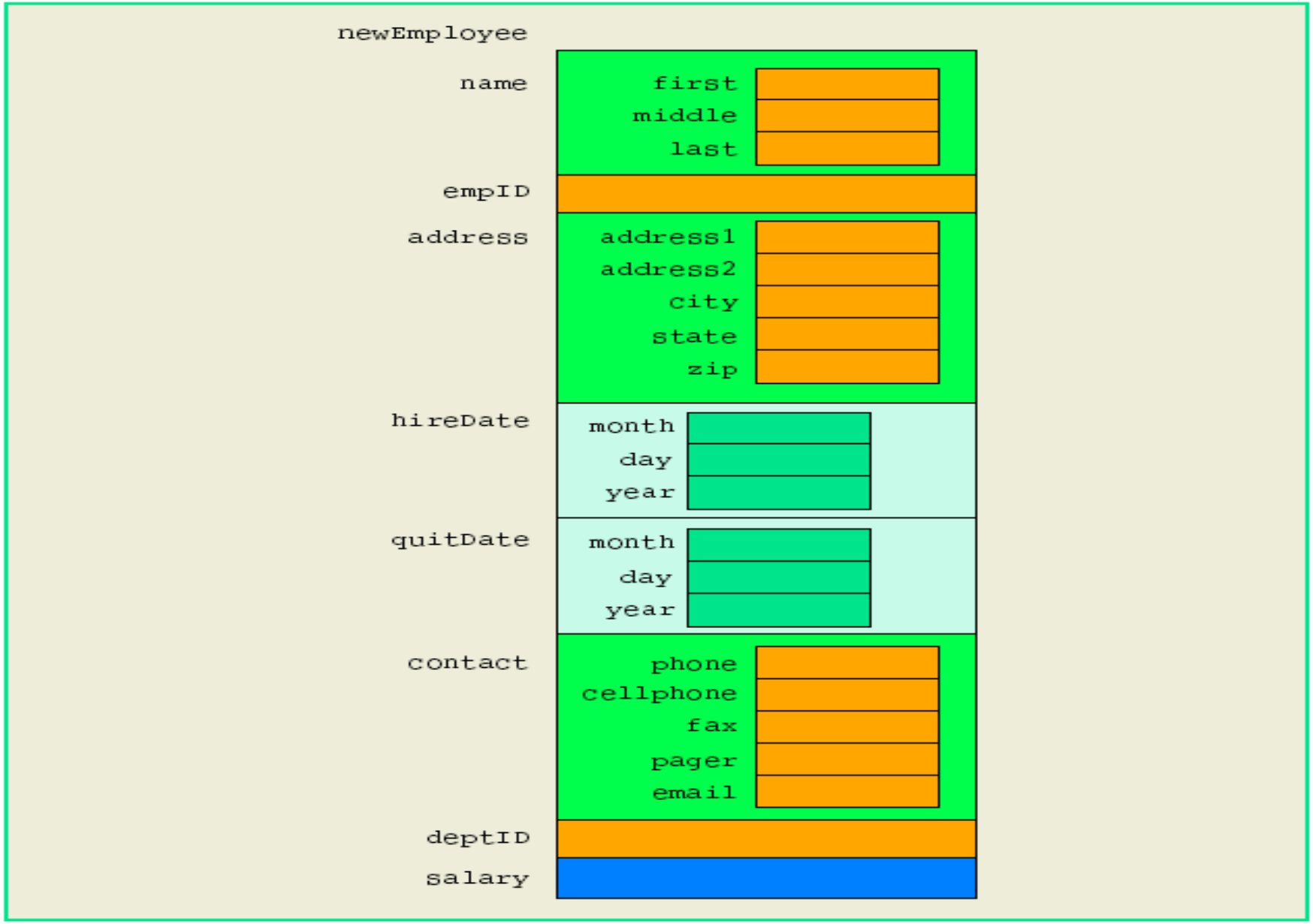
# Arrays in structs

- Two key items are associated with a list:
  - Values (elements)
  - Length of the list
- Define a struct containing both items:

```
const arraySize = 1000;
struct listType
{
  int listElem[arraySize];   //array containing the list
  int listLength;            //length of the list
};
```




Array of employees




struct variable newEmployee

# Summary

- Struct: collection of a fixed number of components
- Components can be of different types
- struct is a reserved word
- No memory is allocated for a struct; memory is allocated for struct variables when declared
- Components of a struct are called members

# Summary (cont.)

- struct components are accessed by name
- Dot (.) operator is called the member access operator
- Members of a struct are accessed using the dot (.) operator
- The only built-in operations on a struct are the assignment and member access

# Summary (cont.)

- Neither arithmetic nor relational operations are allowed on the entire structure
- structures can be passed by value or reference
- A function can return a structure
- A structure can be a member of another structure

# List ADT

A general list of size N, of the form: $A0, A1, A2, ..., AN-1$.

N=0 -> empty list

For any non-empty list:

$Ai$ follows (or succeeds) $Ai-1$ ($i < N$) and

$Ai-1$ precedes $Ai$ ($i > 0$)

We will not define the predecessor of $A0$ or the successor of $AN-1$

The **position** of element $A_i$ in a list is $i$

The first element of the list is $A0$, and the last element is $AN-1$

# List ADT
# Set of Operations

- Insert
- Remove
- Find
- Next
- Previous
- Print
- Clear

# List ADT
# Set of Operations

Given list: 34, 12, 52, 16, 12

Find(52) -> returns 2

Insert(25, 2) -> 34, 12, 25, 52, 16, 12

Remove(52) -> 34, 12, 15, 16, 12

Next(2) -> 16 // next of given index

Previous(2) -> 12 // prev of given index

What does Find(8) returns?

The interpretation of what is appropriate for a function is entirely up to the programmer, as is the handling of special cases.

# A Simple Array Implementation of Lists

```
// Array-based list implementation
class AList : public List {
  ListItemType* listArray;            // Array holding list elements
  static const int DEFAULT_SIZE = 10; // Default size
  int maxSize;                        // Maximum size of list
  int listSize;                       // Current # of list items
  int curr;                           // Position of current element
```

Ref: OpenDSA Project
https://opendsa-server.cs.vt.edu/ODSA/Books/Everything/html/ListArray.html

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```

```
public:
  // Constructors
  // Create a new list object with maximum size "size"
  AList(int size = DEFAULT_SIZE) : listSize(0), curr(0) {
    maxSize = size;
    listArray = new ListItemType[size];         // Create listArray
  }

  ~AList() { delete [] listArray; }      // destructor to remove array
```

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```

```
  public:
```

```
    // Reinitialize the list
    void clear() { listSize = curr = 0; }  // Simply reinitialize values
```

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  int curr;                             // Position of current element
```

```
public:
```

```
// Insert "it" at current position
bool insert(const ListItemType& it) {
  if (listSize >= maxSize) return false;
  for (int i = listSize; i > curr; i--)  // Shift elements up
    listArray[i] = listArray[i-1];       // to make room
  listArray[curr] = it;
  listSize++;                            // Increment list size
  return true;
}
```

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```
  public:
```

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  int curr;                             // Position of current element
```

```
public:
```

```
// Remove and return the current element
ListItemType remove() {
  if ((curr < 0) || (curr >= listSize)) // No current element
    throw std::out_of_range("remove() in AList has current of " + to_string(curr) + " and size of "
      + to_string(listSize) + " that is not a a valid element");
  ListItemType it = listArray[curr];     // Copy the element
  for(int i = curr; i < listSize-1; i++) // Shift them down
    listArray[i] = listArray[i+1];
  listSize--;                            // Decrement size
  return it;
}
```

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  int curr;                           // Position of current element
```

```
public:
```

```
// Set current list position to "pos"
bool moveToPos(int pos) {
  if ((pos < 0) || (pos > listSize)) return false;
  curr = pos;
  return true;
}
```

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```

```
public:
```

```
void moveToStart() { curr = 0; }        // Set to front
void moveToEnd() { curr = listSize; }   // Set at end
void prev() { if (curr != 0) curr--; } // Move left
void next() { if (curr < listSize) curr++; } // Move right
int length() { return listSize; }       // Return list size
int currPos() { return curr; }          // Return current position
```

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```

```
public:
```

```
// Return true if current position is at end of the list
bool isAtEnd() { return curr == listSize; }
```

```
// Check if the list is empty
bool isEmpty() { return listSize == 0; }
```

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  int listSize;                         // Current # of list items
  int curr;                             // Position of current element
```

```
public:
```

```
// Return the current element
ListItemType getValue() {
  if ((curr < 0) || (curr >= listSize)) // No current element
    throw std::out_of_range("getvalue() in AList has current of " + to_string(curr) +  + " and size of "
      + to_string(listSize) + " that is not a a valid element");
  return listArray[curr];
}
```

Ref: OpenDSA Project
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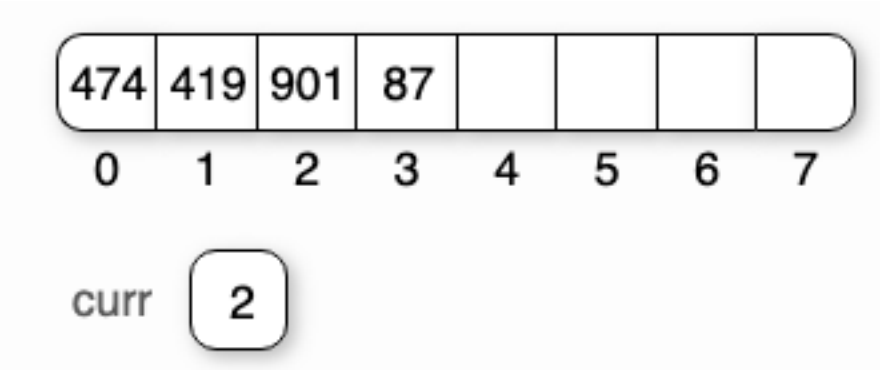
# Insert

```
// Insert "it" at current position
public boolean insert(Object it) {
  if (listSize >= maxSize) return false;
  for (int i=listSize; i>curr; i--)  // Shift elements up
    listArray[i] = listArray[i-1];   //   to make room
  listArray[curr] = it;
  listSize++;                        // Increment list size
  return true;
}
```

insert 512 to current position

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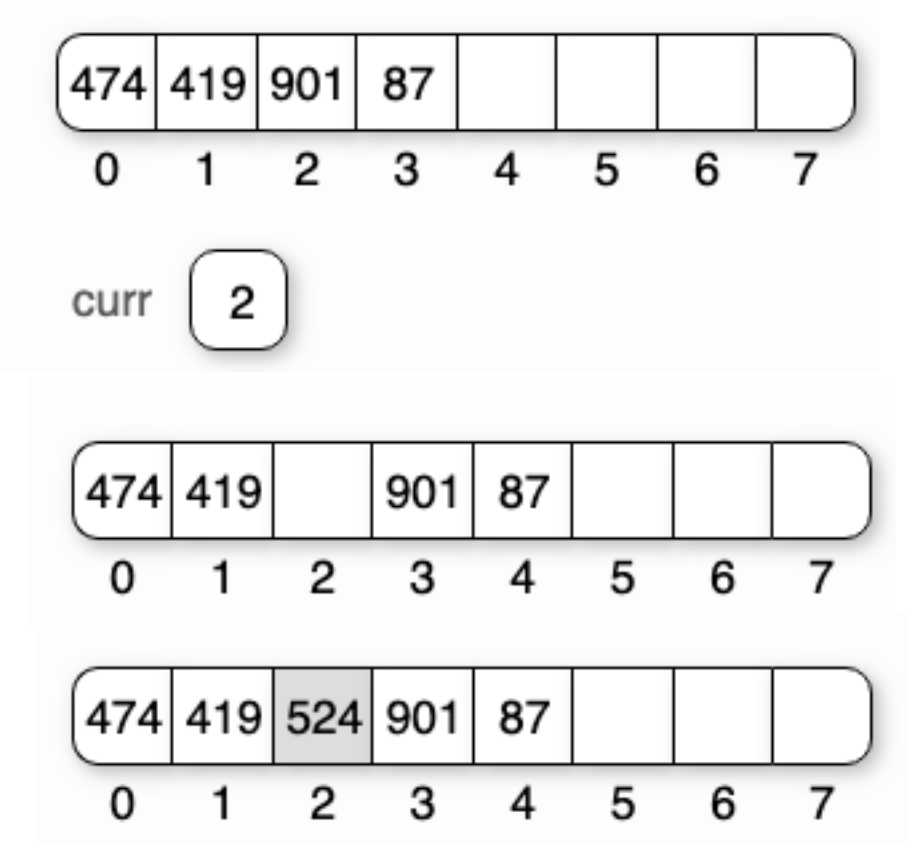
    listArray[i] = listArray[i-1];   //   to make room
  listArray[curr] = it;
  listSize++;                        // Increment list size
  return true;
}
```

insert 524 to current position

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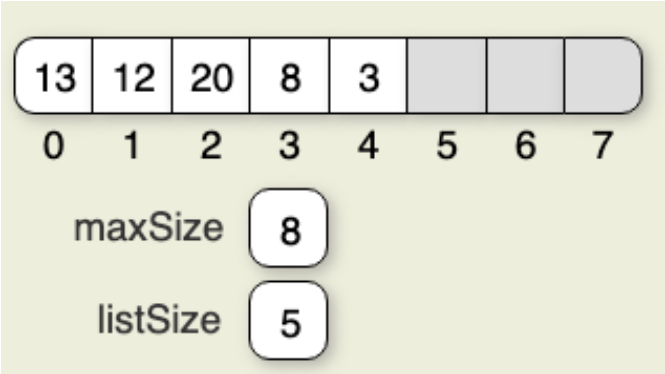
# Append

Append 23

```
// Append "it" to list
public boolean append(Object it) {
  if (listSize >= maxSize) return false;
  listArray[listSize++] = it;
  return true;
}
```

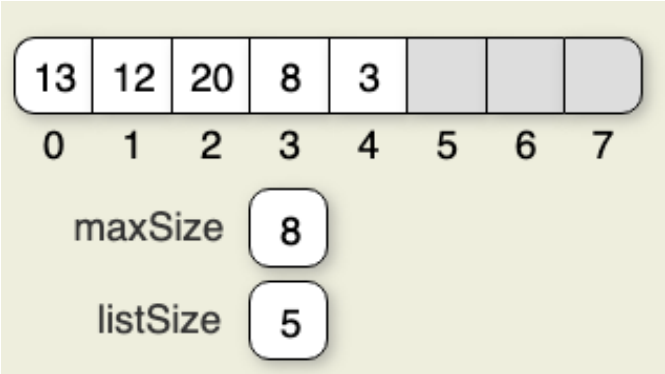


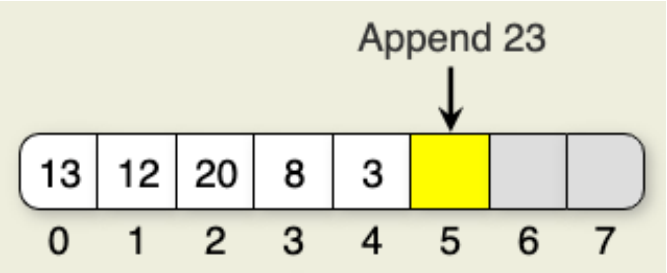

Ref: OpenDSA Project
https://opendsa-server.cs.vt.edu/ODSA/Books/Everything/html/ListArray.html

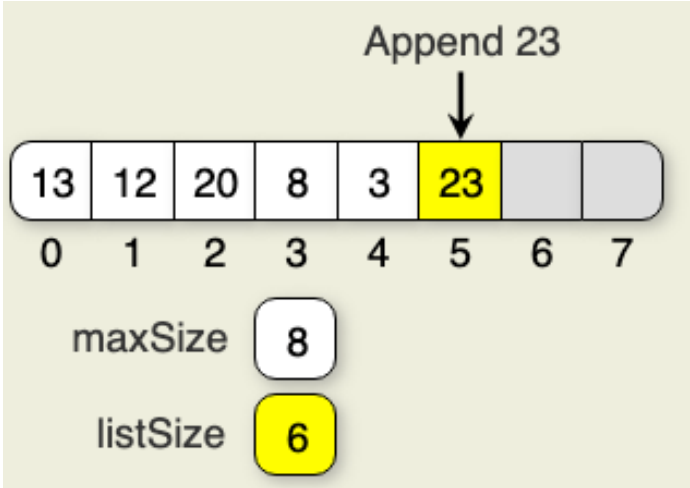
# Append

Append 23

```
// Append "it" to list
public boolean append(Object it) {
  if (listSize >= maxSize) return false;
  listArray[listSize++] = it;
  return true;
}
```

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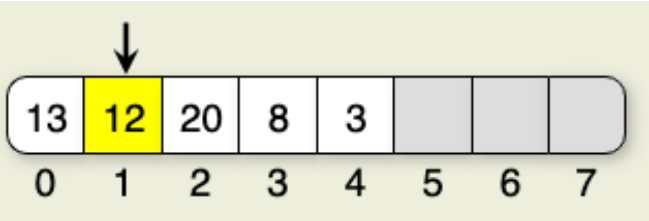
# Remove

Remove 12 in position 1

```
// Remove and return the current element
public Object remove() throws NoSuchElementException {
  if ((curr<0) || (curr>=listSize))  // No current eleme
    throw new NoSuchElementException("remove() in AList
      + listSize + " that is not a a valid element");
  Object it = listArray[curr];       // Copy the element
  for(int i=curr; i<listSize-1; i++) // Shift them down
    listArray[i] = listArray[i+1];
  listSize--;                        // Decrement size
  return it;
}
```

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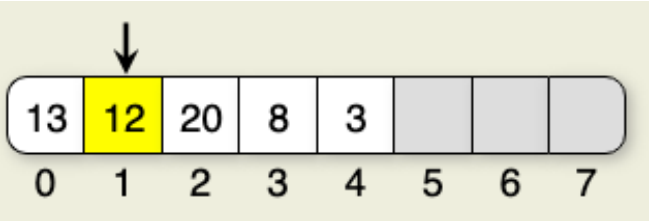


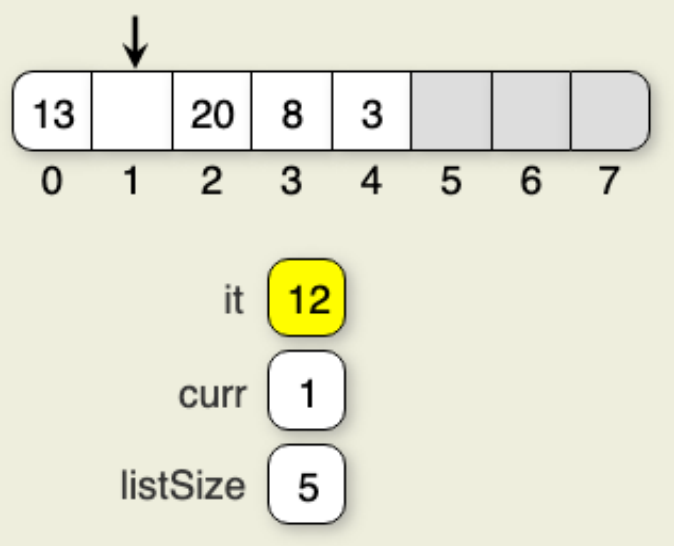


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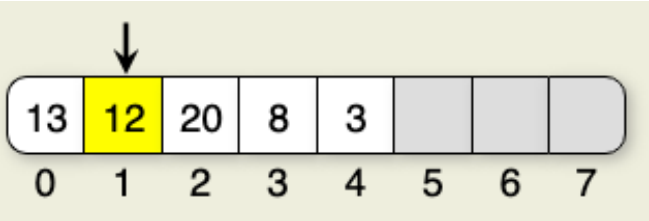


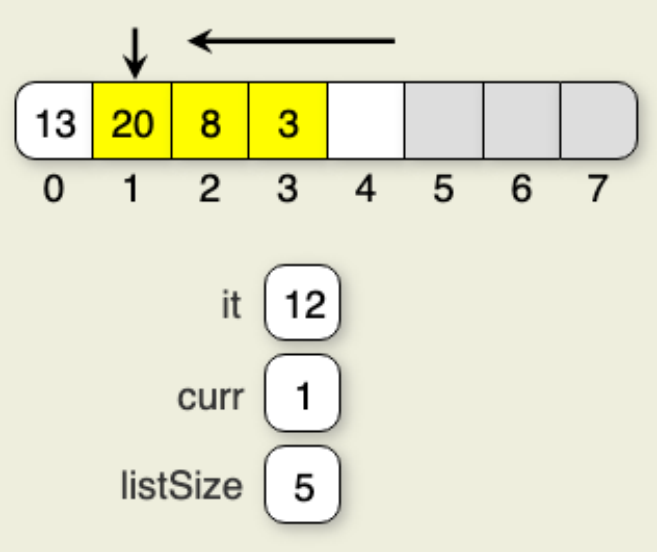


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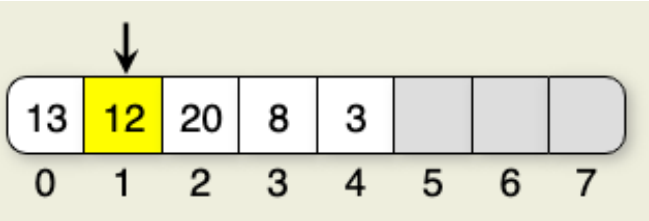


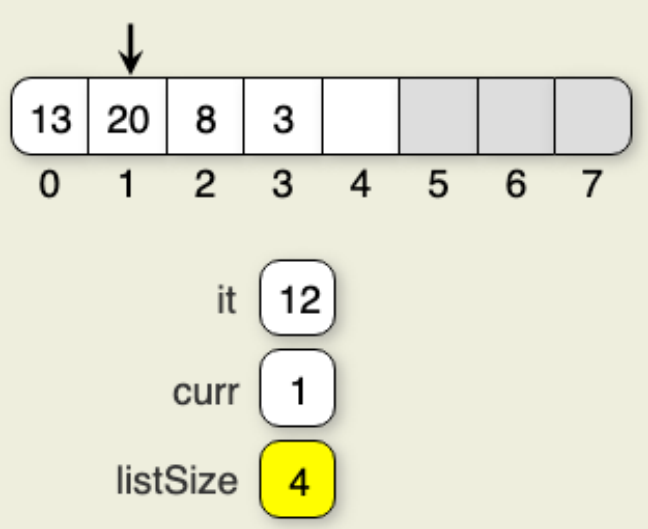


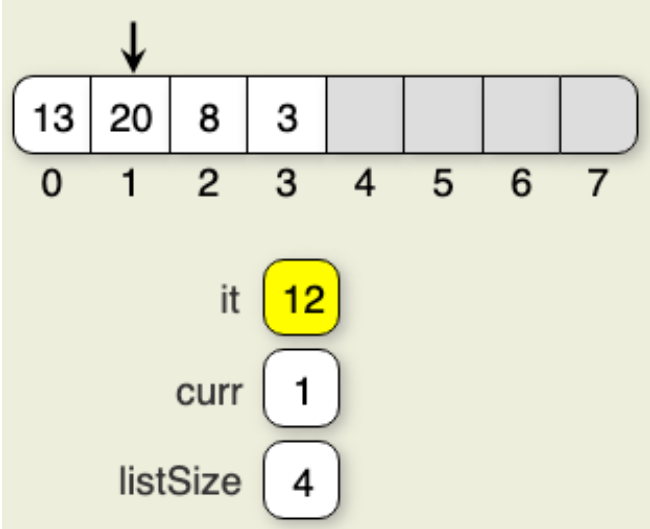


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# Computational Times

Printing list -> linear time
Empty list -> constant time
Insert & remove ops -> expensive

- depends on where the insertions and deletions occur

# Computational Times

## insertion (worst case)

- Insert at the front of the list
- What is the complexity?

## insertion (best case)

- Insert at the end of the list
- What is the complexity?

# Computational Times

## Remove (worst case)

- Deleting the first element of the list
- What is the complexity?

## Remove (best case)

- Deleting the last element of the list
- What is the complexity?

# Remarks

There are many situations where the list is built up by insertions at the high end,

and then only array accesses (i.e., getValue operations) occur.

In such a case, the array is a suitable implementation.

if insertions and deletions occur throughout the list and,

in particular, at the front of the list,

then the array is not a good option.